# Nikhil D'Souza

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## Education

**Columbia University** – New York City, NY **Dec 2023**

- M.S. in Data Science
- Coursework: Probability & Statistics, Applied Machine Learning, Math of Finance, Finance & Structuring for Data Science

**Purdue University** – West Lafayette, IN **May 2022**

- B.S. in Computer Science, Data Science, Applied Statistics (GPA: 3.84)
- Coursework: Advanced Neural Projects, Data Mining & ML, Analysis of Algorithms, Neural Image Processing, Intro to AI, Data Structures & Algorithms, Large-Scale Data Analytics, Regression Analysis, Probability, Linear Algebra, Multivariable Calculus
- Head Teaching Assistant of The Data Mine (1000+ students and 50+ corporate partners)

**Coursera** – deeplearning.ai by Andrew Ng **2020**

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization

## Experience

**Vitalize** – Co-founder & CTO **Feb - Sep 2022**

- Built the first digital wellness platform for the healthcare workforce (vitalizecare.co).
- Developed a clinician-centered mobile app and web-based admin dashboard using MongoDB, Express.js, React, and Node.js.
- Led development team to build anonymous video coaching, dynamic surveys, and analytics dashboard.

**Fast MRI Reconstruction using Deep Learning** – Undergraduate Researcher **Aug 2021 - May 2022**

- Investigated the use of AI to make MRI scans up to 10x faster by reconstructing accurate images from undersampled data.
- Trained a generative adversarial network with images that are Fourier transforms of 20% subsampled and fully sampled *k*-space data.
- Utilized a GAN with a U-Net-based generator architecture and residual-in-residual dense blocks (RRDBs).

**Eli Lilly** – AI/ML Intern **May - Aug 2021**

- Analyzed clinical trial data to predict trial site performance and identify new indications of in-market Lilly drugs.
- Used PCA, logistic regression, and XGBoost to determine if Lilly-sponsored trials should continue being funded.
- Identified reasons why investigators reforecast trial enrollment estimates using TF-IDF.

**Vincere Health** – Software Engineer & Data Scientist **Oct 2019 - Aug 2020**

- Built a platform that uses financial incentives, behavioral nudges, and evidence-based interventions to help quit smoking and be healthy.
- Developed patient facial recognition, chat messaging using sockets, and incentive engine using Redis.
- Spearheaded data science capability providing actionable insights, user performance, and patient rewards.

**Saama** – AI Intern **Jun - Aug 2018**

- Used supervised ML techniques to identify packaging anomalies on a conveyor belt for a leading biopharmaceutical company.
- Developed CNN with 99% validation accuracy and leveraged image augmentation to expand the dataset.

## Projects and Awards

**Mask Autoencoder** – Smart Software Hackathon (2nd Place Overall Winner) **2021**
**Feverbase** – COVID clinical trial data platform **2020**
**Harvard President's Innovation Challenge** – Grand Prize Winner **2020**
**L3Harris Technologies Scholarship** – Purdue CS Awards **2019**
**Calculating Fatigue of Rugby Athletes** – ASA DataFest (1st Place Overall Winner) **2019**
**Predicting Emerging eSport Celebrities** – Krannert eSport Hackathon (3rd Place Overall Winner) **2018**
**GluClose** – PennApps XVIII (2nd Place Overall Winner) **2018**
**Apple Worldwide Developer Conference Scholarship Winner** **2015**

## Skills

**Languages/Tools:** Python, R, SQL, Tensorflow, Pytorch, Flask, React, Typescript, JS ES6, Node.js, Swift, Java, C, Unix, Git, Docker, Redis
**AWS:** EC2, S3, Lambda, API Gateway, Sagemaker, Elasticsearch, CloudWatch, Rekognition, Comprehend, SNS, Cognito